

ALEX MORGAN

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PROFESSIONAL SUMMARY

Senior Machine Learning & Full-Stack Engineer with 5+ years of experience designing and deploying scalable deep learning architectures, microservices, and distributed cloud applications. Specialized in Computer Vision, PyTorch, FastAPI, and PostgreSQL.

TECHNICAL SKILLS

- Languages: Python, TypeScript, Go, SQL, C++
- Frameworks & Libraries: PyTorch, TensorFlow, FastAPI, Next.js, React, SQLAlchemy
- Deep Learning: CNNs, Vision Transformers (ViT), ResNet, Transfer Learning, Model Quantization
- Cloud & DevOps: Docker, Kubernetes, AWS (S3, EC2), PostgreSQL, Redis, CI/CD

EXPERIENCE

Senior AI Engineer | NeuralScale Technologies (2022 - Present)

- Architected and deployed an end-to-end computer vision classification microservice handling 4M+ daily API requests with sub-100ms P99 latency using FastAPI and Redis.
- Led model optimization and TensorRT quantization, reducing GPU inference memory footprint by 42%.
- Mentored 4 junior engineers on distributed training workflows, ML test coverage, and Docker containerization.

Software Engineer | Apex Systems (2019 - 2022)

- Built enterprise RESTful APIs using Python, FastAPI, and PostgreSQL with SQLAlchemy 2.0 and Alembic.
- Implemented real-time telemetry and audio processing pipelines using WebSockets and WebRTC.
- Integrated automated CI/CD test runners with Pytest, cutting deployment cycle times by 35%.

KEY PROJECTS

Coral Reef Health Classification with CNN & Vision Transformers (ViT)

- Designed a hybrid deep learning model combining ResNet-50 CNN backbones with a Vision Transformer encoder to classify coral degradation from underwater drone imagery.
- Achieved a 94.6% F1-score across 8 coral health categories on a dataset of 35,000 high-resolution aerial and subsea images.
- Implemented data augmentation pipelines using Albumentations and trained models with PyTorch Lightning on multi-GPU AWS EC2 instances.

Distributed Real-Time Audio Transcription Microservice

- Created a scalable real-time voice streaming system using WebRTC, FastAPI, and Whisper.
- Deployed on Kubernetes with automatic horizontal pod autoscaling based on active voice connection streams.

EDUCATION

Bachelor of Science in Computer Science

University of California, Berkeley (2015 - 2019)

- Honors: Dean List, Focus on Artificial Intelligence & Distributed Systems

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- DeepLearning.AI Deep Learning Specialization